KPIT APEX LAB SUMMER INTERNSHIP 26

Intelligent School Bus Safety System

Retrofit Traffic Sign Detection & Automatic Speed Compliance

DEVELOPED BY

Tanishka Ubhe, Anushka Joshi, Aditi Patil

PROJECT GUIDE

Abhinivesh Gunasekaran

The School Bus Safety Gap

Safety Challenge

School buses operate daily in crowded, high-traffic urban environments where driver distraction, overspeeding, or obscured signs pose major hazards.

Human operational error remains the leading factor in road accidents. Drivers face high cognitive load keeping watch on children and the road simultaneously.

Retrofit Need

The vast majority of existing, active school bus fleets lack modern Advanced Driver Assistance Systems (ADAS) or intelligent crash-avoidance aids.

Procuring new buses integrated with factory-installed safety technologies is financially prohibitive. A modular, low-cost retrofit platform is critically required.

YOLO Traffic Sign Detection

Dataset & Target Classes

Diverse traffic sign imagery was aggregated and configured to build an accurate training footprint for active warning signals.

- **Sourcing:** Curated files from Kaggle and Roboflow databases.
- **Targets:** High-priority focus on Speed Limits, Stops, and Light signs.
- **Processing:** Converted GTSDB datasets into YOLO format.

Real-Time Inference Pipeline

An optimized edge computer vision pipeline detects, validates, and acts on physical road conditions continuously.

- **Acquisition:** Pi Camera capturing frame arrays at driver's view.
- **Model Run:** Lightweight Ultralytics YOLOv8 running in real time.
- **CAN Transit:** Detected classes sent over MCP2515 CAN Bus.

Study of Academic Literature

Detection Pipelines

Traditional image processing (Gaussian filtering, contour parsing, edge detection) helps define bounding boundaries and reduce processing window sizes on embedded CPUs.

Deep Learning Superiority

Research proves Convolutional Neural Network (CNN) models yield far superior classification stability and weather robustness compared to simple color-based thresholding.

Environmental Handling

Modular shape-based grayscale analysis models help secure robust identification despite intense shadow transitions, motion blur, and tree obstructions on roads.

Hardware Optimization

Literatures examining edge-embedded GPU and Single-Board Computer architectures guided the implementation of YOLO running natively on modern Raspberry Pi boards.

Integrated Hardware Design

Primary Processor

Raspberry Pi 5 serves as the system's high-speed AI engine. It captures video arrays from the CSI camera ribbon and processes YOLO model inference frames in real time.

Control & Actuation

STM32 Nucleo-F446RE manages low-level operations. It reads pulse periods from the Hall sensor, runs logic, and drives dual H-Bridge controllers.

Automotive CAN Bus

MCP2515 SPI Modules bridge both systems. They establish an error-free, automotive-grade CAN link to securely stream real-time speed data packets.

Complete Software Ecosystem

System Domain	Software & Frameworks	System Role & Capabilities
AI & Computer Vision	Raspberry Pi OS, OpenCV, Ultralytics YOLO	Handles real-time image acquisition, pre-processing, class prediction, and bounding box rendering.
Embedded Firmware	Arduino IDE, STM32CubeMX, C/C++	Executes high-frequency timer interrupts, measures pulse sensor frequency, and generates PWM outputs.
Inter-system Link	MCP2515 CAN protocol library, Python-CAN	Formats detection messages into robust CAN packets, managing error-free data sync at high baud rates.
Telemetry Dashboard	Streamlit, Power BI, Python logs	Generates CSV logs of speed, sign data, and overrides; sends live telemetry over Wi-Fi for remote viewing.

Automatic Speed Compliance

Pulse-Driven Speed Monitoring

Instead of manual velocity checking, the STM32 handles non-contact speed monitoring dynamically over high-frequency hardware pins.

- **A3144 Hall Sensor:** Generates digital ticks as wheel magnets pass.
- **Timer Interrupts:** Measures pulse count per designated interval.
- **RPM Calculations:** Transcribes pulse delta periods into wheel RPM.

INTERRUPT PRIORITY

Processed at microsecond thresholds to avoid missed counts during high vehicle velocities.

Closed-Loop Throttle Compliance

The system regulates school bus velocity through a closed feedback mechanism, prioritizing child safety without driver override.

- **Dynamic Evaluation:** Continually cross-references current RPM.
- **PWM Duty Scaling:** Reduces L293D motor power output instantly.
- **Forced Halt:** Zeroes duty cycles in Red Light or Stop states.

FAIL-SAFE ACTION

In case of CAN communication loss, motors default to 20% safe speed thresholds automatically.

Six-Week Project Timeline

Week 1-2

Research & Setup

OS flashing, STM32 Cube validation, Pi camera ribbon configuration, and basic UART interface debugging.

Week 3

Model Training

YOLO training on GTSDB datasets over 3[illegible] epochs, managing thermal limits of Raspberry Pi 5 under heavy workloads.

Week 4-5

CAN & Controls

MCP2515 SPI interface validation, Hall speed calibration, closed-loop PWM regulation, and Streamlit dashboard integration.

Week 6

System Assembly

3D-printed chassis assembly, independent 4-motor soldering, backup CSV telemetry update, and battery-powered run tests.

Core Challenges & Solutions

VCC & CAN Integrity

The Challenge: Early multi-node integration suffered from corrupted CAN frames and STM32 locks because nodes shared a common VCC rail, causing signal ground shifts and voltage drops.

The Solution: Power domains were isolated by separating the high-current motor supply from the logic supplies while keeping a strict common signal ground plane, securing 100% CAN frame delivery.

Thermals & Mechanical Noise

The Challenge: Sustained AI YOLO inference pushed the CPU to critical limits causing rapid thermal throttling. Simultaneously, mechanical wheel wobble created erratic Hall-effect pulses.

The Solution: Implemented active heatsink cooling fans on the Pi 5. Fabricated a precision secondary calibration card to stabilize the sensor array, providing jitter-free RPM outputs.

Real-Time Telemetry Dashboard

Power BI Bottlenecks

Early iterations utilized Power BI connected to asynchronous CSV dumps to visualize road conditions. This introduced severe limitations:

- **Sync Delays:** Cloud-database write refresh latencies.
- **Lag Spike:** Intermittent delays displaying speed changes.
- **No Live Action:** Incapable of continuous, sub-second telemetry.

Streamlit Web Solution

An alternative, lightweight Python web-telemetry system was engineered to run locally and display real-time variables.

- **Local Processing:** Direct asynchronous file updates.
- **Wi-Fi Transit:** Real-time transfer to remote fleet observer laptop.
- **Accurate Logs:** Instantly captures speed and sign records.

System Assembly & Validation

The 3D-Printed Bus Prototype

The final validation platform integrated the full electronics stack onto a custom, robust 3D-printed bus frame designed by the mentor to prove the retrofit deployment potential.

By soldering independent 4-motor lines and integrating dual H-Bridge controllers, the drive train successfully demonstrates automatic velocity override, emergency stop compliance, and telemetry logging.

Battery & Embedded Power

Tested successfully under independent, onboard lead-acid battery power. This transition from standard laboratory power supplies proved the prototype's standalone operational capacity.

Electrical and sensor lines are securely soldered onto a centralized bus platform, avoiding mechanical disconnects or noise spikes during rugged movement runs.

Conclusion & Future Work

An Affordable Retrofit ADAS Platform for School Bus Fleets

Summary of Accomplishments: Real-time YOLO-based traffic sign detection on Raspberry Pi 5 coupled with STM32-based PWM closed-loop motor compliance control, connected seamlessly over a robust dual-node MCP2515 CAN bus network.

Future Road Map: Integration of high-sensitivity GPS modules for live fleet geo-fencing, cellular cloud logging integration, driver fatigue tracking, and native ECU/OBD-II CAN bus throttling.

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